

JACOB MILLER

iOS Developer | Swift • SwiftUI • UIKit • Core Data | 4+ Years Building & Shipping iOS Applications

EDUCATION

McHenry County College

Crystal Lake, IL

- Associate's in Mobile App Development (2019–2023)
- Associate's in Web Development (2019–2024)
- Named on the President's Honor List, 2023 Spring Semester.

DETAILS

Johnsburg, IL, United States

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LINKS

[Portfolio Website](#)

[LinkedIn](#)

[GitHub](#)

iOS SKILLS

Swift

UIKit

SwiftUI

Swift Charts

Core Data

Xcode

CocoaPods

DESIGN PATTERNS

MVC

MVVM

ADDITIONAL SKILLS

Firebase

REST APIs

Git

JSON

PROFESSIONAL SUMMARY

iOS developer with an Associate's in Mobile App Development and 4+ years designing, shipping, and maintaining production Swift applications. Published and actively maintain Iron Journal, an App Store fitness application built with Swift, UIKit, SwiftUI, and Core Data. Focused on building performant, maintainable applications with clean architecture and intuitive user experiences.

★ PROJECTS & EXPERIENCE (4+ YEARS)

Iron Journal (Published iOS App)

Structured Weight Training & Progress Tracker

March 2022 — Present (4+ years)

- Architected a **phase-driven workout engine** supporting dynamic exercise ordering, mid-session edits, backward session navigation with preserved inputs, and full workout recovery after app termination.
 - Designed a **hierarchical Core Data model** linking routines, workouts, goals, and historical performance, enabling fast access across 3-month, 6-month, 1-year, and all-time ranges.
 - Engineered a workout-wide **rest retrieval algorithm** that reconstructs accurate pre- and post-set rest intervals regardless of exercise order, enabling reliable live tracking and historical workout analysis.
 - Developed a **real-time progression algorithm** that dynamically recalculates prescribed weights and reps from live workout performance while preserving the original workout plan, with progression logic validated through **unit tests**.
 - Maintained a **hybrid UIKit + SwiftUI architecture** with strict MVVM boundaries and multithreaded Core Data operations to ensure UI responsiveness during heavy data access.
 - Available on the App Store: [Iron Journal](#)
- **Tech Stack:** Swift, UIKit, SwiftUI, Swift Charts, Core Data, Firebase, MVVM

Custom REST APIs (PHP/MySQL)

Backend infrastructure built and deployed for [JacobJMiller.com](#).

February 2025 — Present

- Built a RESTful email-sending API (PHP, PHPMailer) with a custom token-bucket rate limiter using file locking and per-IP JSON tracking, returning structured JSON for real-time UI feedback and error handling
 - Designed a lightweight RESTful API for full CRUD operations on MySQL-backed user and usage data, replacing a prior Node.js/Express version to cut hosting costs and simplify maintenance
 - Deployed both APIs to production for use across JacobJMiller.com and the companion iOS Dashboard App
- **Tech Stack:** PHP, MySQL, PHPMailer, Apache (managed via cPanel)
- **GitHub Projects:** [Email-API-PHP](#), [Mysql-API-PHP](#)

iOS-Dashboard-App (Demo Project)

Designed to demonstrate data visualization and database sharing between different platforms with Swift Charts.

September 2024

- Designed and deployed a custom RESTful API (PHP/MySQL) with full CRUD support powering the client.
 - Architected the backend to share a single database with a companion web dashboard, syncing metrics across both platforms
- **Tech Stack:** Swift, SwiftUI, Swift Charts, MVVM